

Q1:- A particular x-ray photon has a wavelength of 41.6 pm. Calculate the photon's energy, frequency and momentum.

As we know:-

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{(41.6 \times 10^{-12} \text{ m})}$$

$$E = 4.7 \times 10^{-15} \text{ J}$$

$$E = 29864.7 \text{ eV}$$

$$\text{Frequency} = \frac{c}{\lambda} = \frac{3 \times 10^8}{41.6 \times 10^{-12} \text{ m}} = 7.21 \times 10^{18} \text{ s}^{-1}$$

$$P = \frac{E}{c} = \frac{29864.7}{3 \times 10^8} = 9.95 \times 10^{-5} \text{ eV/ans}$$

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Q2:- Find (a) frequency, (b) wavelength and (c) the momentum of photon whose energy equals rest energy of electron.

(a) Rest energy of electron = 0.5 MeV
 $= 8 \times 10^{-14} \text{ J}$

$$\text{Frequency} = \frac{E}{h}$$

$$= \frac{8 \times 10^{-14} \text{ J}}{6.626 \times 10^{-34} \text{ Js}} = \frac{8 \times 10^{-14} \text{ J}}{6.626 \times 10^{-34} \text{ Js}}$$

$$\boxed{f = 1.20 \times 10^{20} \text{ s}^{-1}}$$

(b) $f = \frac{c}{\lambda} \Rightarrow \lambda = \frac{c}{f}$

$$\lambda = \frac{3 \times 10^8}{1.20 \times 10^{20}} = 2.43 \times 10^{-12} \text{ m}$$

$$\boxed{\lambda = 2.43 \text{ pm}}$$

(c)

$$\text{momentum} = \frac{E}{c}$$

$$= \frac{0.5 \text{ MeV}}{c}$$

$$\boxed{p = 0.511 \times 10^6 \text{ eV}/c}$$

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Q3:- By how much does a sodium atom slow down on absorbing a photon of wavelength 589 nm with which it collides head on?

Initial momentum of the system is momentum of photon

$$\text{which is } p = \frac{E}{c} = \frac{hf}{c} = \frac{hf}{\lambda c}$$

$$p = h/\lambda$$

final momentum is $p = mv$

$$v = \frac{p}{m} \Rightarrow v = \frac{h/\lambda}{m}$$

$$v = \frac{6.626 \times 10^{-34}}{(589 \times 10^{-9})(9.1 \times 10^{-31})}$$

$$v = 2.9 \text{ cm/s}$$

Q4:- The quantity h/mc is often called Compton wavelength λ_c of scattering particle and that equation is written as

$$\Delta\lambda = \lambda_c (1 - \cos\phi)$$

(a) calculate Compton wavelength of an electron, of a proton.

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mass of electron = 9.1×10^{-31} kg

$$\lambda_c = \frac{h}{mc} = \frac{6.626 \times 10^{-34}}{(9.1 \times 10^{-31})(3 \times 10^8)}$$

$$\lambda_c = 2.43 \text{ pm}$$

mass of proton = 1.67×10^{-27} kg

$$\lambda_c = \frac{h}{mc} = \frac{6.626 \times 10^{-34}}{(1.67 \times 10^{-27})(3 \times 10^8)}$$

$$\lambda_c = 1.32 \times 10^{-15} = 0.00132 \times 10^{-12}$$

$$\lambda_c = 0.0013 \text{ pm}$$

(b) what is energy of photon whose wavelength is equal to the Compton wavelength of electron, of the proton.

$$E = hf = \frac{hc}{\lambda}$$

For electron

$$E = (6.626 \times 10^{-34})(3 \times 10^8)$$

$$2.43 \times 10^{-12} \text{ m}$$

$$E = 8.18 \times 10^{-14} \text{ J}$$

For Proton

$$E = (6.626 \times 10^{-34})(3 \times 10^8)$$

$$1.325 \times 10^{-15}$$

$$E = 1.50 \times 10^{-10} \text{ J}$$

(c) show that in general energy of a photon whose wavelength is equal to Compton wavelength of a particle is same as the rest energy of that particles.

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As:-

$$E = hf = \frac{hc}{\lambda}$$

$$\text{and } \lambda = \frac{h}{mc}$$

$$E = \frac{hc}{\lambda} \cdot mc = mc^2$$

$$\boxed{E = mc^2}$$

Hence, proved.

Q5:- Photons of wavelength 2.17 pm are incident on electrons. Find wavelength of photon that is scattered 35° from incident radiation.

$$\Delta \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$$\lambda' - \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$$\lambda' = \lambda + \frac{h}{mc} (1 - \cos \phi)$$

for electron

$$\left(\frac{h}{mc} = 2.43 \text{ pm} \right)$$

$$= 2.17 \text{ pm} + (2.43 \text{ pm}) (1 - \cos 35^\circ)$$

$$\lambda' = 2.61 \text{ pm}$$

$$(b) \quad \phi = 115^\circ$$

$$\lambda' = \lambda + \frac{h}{mc} (1 - \cos \phi)$$

$$= 2.17 \text{ pm} + 2.43 \text{ pm} (1 - \cos 115^\circ)$$

$$\lambda' = 5.63 \text{ pm}$$

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Q6:- A 511 keV gamma-ray photon is Compton scattered from free electron in an aluminium block (a) what is wavelength of incident photon.

(b) what is wavelength of scattered photon what is energy of scattered photon assuming angle 72° .

(a) $\lambda = \frac{h}{mc} = 2.43 \text{ pm}$

(b) we know that
 $\lambda' = \lambda + \frac{h}{mc} (1 - \cos \phi)$

$$\lambda' = 2.43 \text{ pm} + 2.43 \text{ pm} (1 - \cos 72^\circ)$$

$$\boxed{\lambda' = 4.11 \text{ pm}}$$

(c) $E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34}) (3 \times 10^8)}{(4.11 \times 10^{-12})}$

$$E = 4.83 \times 10^{-16} \text{ J}$$

$$\boxed{E = 301 \text{ keV}}$$

Q7:- show that $\Delta E/E$ is the fractional loss of energy of a photon during a Compton collision is given by:-

$$\frac{\Delta E}{E} = \frac{hf'}{mc^2} (1 - \cos \phi)$$

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As we know:-

$$\lambda' - \lambda = \frac{h}{mc} (1 - \cos \phi) \quad \text{--- (1)}$$

And we also know that

$$E = hf \text{ so}$$

$$\begin{aligned} \Delta E &= E - E' = hf - hf' \\ &= \frac{hc}{\lambda} - \frac{hc}{\lambda'} = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) \end{aligned}$$

$$\Delta E = hc \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right) \quad \text{--- (2)}$$

Putting eq (1) in eq (2)

$$\Delta E = \frac{hc \left(\frac{h}{mc} (1 - \cos \phi) \right)}{\lambda \lambda'}$$

$$\Delta E = \frac{h^2 (1 - \cos \phi)}{m \lambda \lambda'} \quad \text{--- (3)}$$

dividing eq (3) by E ($E = \frac{hc}{\lambda}$).

$$\frac{\Delta E}{E} = \frac{\frac{h^2 (1 - \cos \phi)}{m \lambda \lambda'}}{E}$$

$$= \frac{\frac{h^2 (1 - \cos \phi)}{m \lambda \lambda'}}{\frac{hc}{\lambda}} = \frac{h^2 (1 - \cos \phi)}{m \lambda \lambda'} \times \frac{\lambda}{hc}$$

$$= \frac{h (1 - \cos \phi)}{mc \lambda'}$$

$$\left(\because \lambda' = \frac{c}{f'} \right)$$

$$\frac{\Delta E}{E} = \frac{h (1 - \cos \phi)}{mc \cdot \left(\frac{c}{f'} \right)}$$

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$$\frac{\Delta E}{E} = \frac{hf' (1 - \cos \phi)}{mc^2}$$

mc^2 is rest energy of electron
and hf' is the energy of scattered
photon.

Q38 - What fractional increase in wavelength leads
to a 75% loss of photon energy in a
Compton collision with a free electron?

Ans:-

$$\frac{\Delta E}{E} = 75\% = \frac{3}{4}$$

So:-

$$\begin{aligned} \Delta E &= E - E' = hf - hf' \\ &= \frac{hc}{\lambda} - \frac{hc}{\lambda'} = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) \\ &= hc \left(\frac{\lambda' - \lambda}{\lambda \lambda'} \right) = \frac{hc \cdot \Delta \lambda}{\lambda \lambda'} \end{aligned}$$

$$\frac{\Delta E}{E} = \frac{hc \cdot \Delta \lambda}{\frac{hc}{\lambda} \lambda \lambda'} = \frac{\Delta \lambda}{\lambda'} = \frac{\Delta \lambda}{\lambda + \Delta \lambda}$$

$$\frac{3}{4} = \frac{\Delta \lambda}{\lambda + \Delta \lambda} \Rightarrow 3\lambda + 3\Delta \lambda = 4\Delta \lambda$$

$$\boxed{\Delta \lambda = 3\lambda}$$

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Q9 - Find maximum wavelength shift for Compton collision between a photon and a free proton.

maximum shift occurs when

$$\theta = 180^\circ \quad \text{so,}$$

$$\Delta \lambda_m = \frac{2h}{mc} = \frac{2(6.62 \times 10^{-34})}{(3 \times 10^8)(6.27 \times 10^{-27})}$$

$$\Delta \lambda_m = 7.04 \times 10^{-16} \text{ m}$$

Q10: A 62 Kev X-ray photon falling on a carbon block is scattered by a Compton collision and its frequency is shifted by 0.01%. (a) through what angle does photon scattered? (b) How much K.E. is imparted to the electron.

~~change in wavelength is independent of wavelength so:-~~

$$\Delta \lambda = \frac{h}{mc} (1 - \cos \phi)$$

$E = hf$ frequency shift is identical to energy shift. As we know:-

$$\frac{\Delta E}{E} = \frac{hf'}{mc^2} (1 - \cos \phi)$$

$$0.01\% = \frac{(6.62 \times 10^{-34})(6.2 \text{ Kev})}{(0.511 \text{ MeV})} (1 - \cos \phi)$$

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$$\frac{(0.0001)(0.511 \text{ MeV})}{(6.2 \text{ keV})} = 1 - \cos \phi$$

$$\cos \phi = 1 - 8.24 \times 10^{-9}$$

$$\phi = \cos^{-1}(0.9999999176)$$

$$\phi = 7.3^\circ$$

(b)

$$\frac{\Delta E}{E} = 0.0001 \text{ keV}$$

$$\Delta E = 0.0001 \text{ keV} (E) = (6.2 \text{ keV} \times 0.0001)$$

$$\Delta E = 0.62 \text{ eV}$$

Q11:- calculate fractional change in energy for Compton collision with ϕ equal to 90° for radiation in

(a) microwave range with $\lambda = 3 \text{ cm}$.

for $\phi = 90^\circ$

$$\Delta \lambda = \frac{h}{mc} \text{ then}$$

$$\frac{\Delta E}{E} = \frac{E - E'}{E} = 1 - \frac{E'}{E}$$

$$= 1 - \frac{hf'}{hf} = 1 - \frac{hc/\lambda'}{hc/\lambda}$$

$$= 1 - \frac{\lambda}{\lambda'} \Rightarrow 1 - \frac{\lambda}{\lambda + \Delta \lambda}$$

$$= \frac{h/mc}{\lambda + h/mc} = \frac{\lambda + \Delta \lambda - \lambda}{\lambda + \Delta \lambda}$$

$$\frac{\Delta \lambda}{\lambda + \Delta \lambda} = \frac{h/mc}{\lambda + h/mc}$$

(a) $\lambda = 3 \text{ cm}$

$$\frac{\Delta E}{E} = \frac{2.43 \text{ pm}}{3 \text{ cm} + 2.43 \text{ pm}} = 8.1 \times 10^{-11}$$

(b) $\lambda = 5 \text{ nm}$

$$\frac{\Delta E}{E} = \frac{2.43 \text{ pm}}{5 \text{ nm} + 2.43 \text{ pm}} = 4.86 \times 10^{-6}$$

(c) $\lambda = 0.1 \text{ nm}$

$$\frac{\Delta E}{E} = \frac{2.43 \text{ pm}}{0.1 \text{ nm} + 2.43 \text{ pm}} = 0.0237$$

(d) $\frac{\Delta E}{E} = \frac{2.43 \text{ pm}}{1.80 \text{ pm} + 2.43 \text{ pm}} = 0.651$

Q12:- Through what angle must a 215 keV photon be scattered by a free electron so that it loses 10% of its energy.

$$\frac{\Delta E}{E} = 10\% = 0.1$$

$$0.1 = \frac{hf' (1 - \cos \phi)}{mc^2} \Rightarrow \frac{(0.1)(215 \text{ keV})}{(0.511 \text{ MeV})} (1 - \cos \phi)$$

$$0.1 = 0.3786 (1 - \cos \phi) \Rightarrow \frac{0.1}{0.3786} = 1 - \cos \phi$$

$$\cos \phi = 1 - 0.2640$$

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$$\cos \phi = 0.735$$

$$\phi = \cos^{-1}(0.7359)$$

$$\phi = 42.6^\circ$$

